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NAME: Sohail Ansari STD.: _____ SEC.: _____ ROLL NO.: _____ SUB.: ~~English~~ ~~Maths~~

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Lecture : 01

• Introduction

• Semiconductor Physics

- General concept

- Doping

Basic Electrical Components : R, L, C.

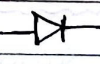
Basic Electronic Component : Diode, Transistor, OPAMP.



Based on semiconductor
(semiconductor physics)

Course Outline :

- Semiconductor Physics (BASIC)

- Diode & Its Modeling $\rightarrow$ 

- Diode Circuits.

- Transistor & Its Modeling $\rightarrow$ 

- Transistor Circuits

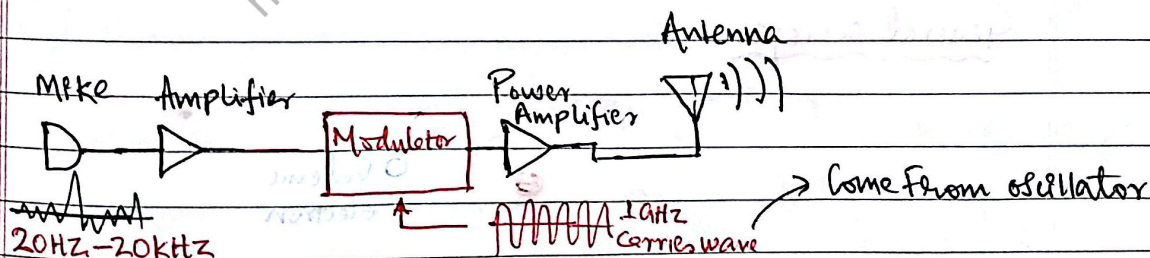
- Mos Transistor $\rightarrow$ 

- Mos Circuits

- OPAMP based Circuits.

We will also look on
Application of electronics

How to build wireless Transmitter?



→ Antenna takes electrical signal & converts into electromagnetic signal so that the waves can propagate through air & get to the receiver

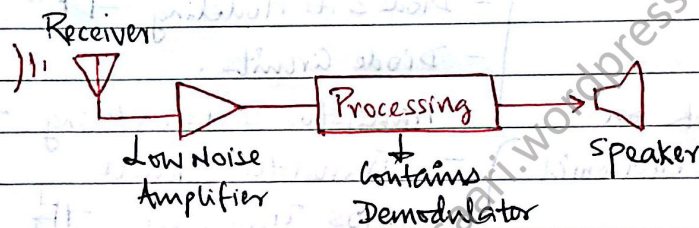
→ To Transmit the signal through Antenna, its dimensions (round/rectangular) must be match with the wavelength of signal that we applied to it. Audio signal range is b/w 20KHz-20KHz (means wavelength is extremely large). So, we

do not connect ~~MPC~~ ^{MPC} directly to Antenna.

→ Therefore to transmit signal we have to increase the Frequency in order to reduce the wavelength by superimposing high Frequency Carrier wave with the actual signal. This is done by Modulator

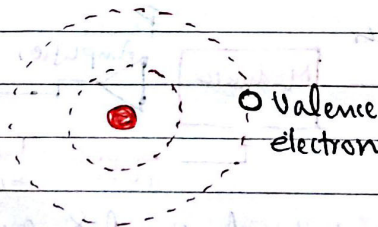
→ Now, Antenna takes ~~large~~ power to transmit signal. So a power amplifier is required.

wireless Receiver



Let's Begin with Semiconductor Physics

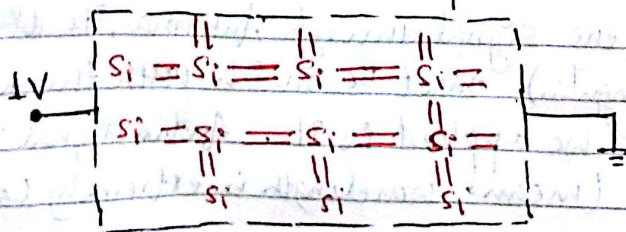
General concepts :



Sodium(Na) : 1 Valence electrons (Very Reactive)

Neon : 8 valence electrons (Non-Reactive)

Silicon : 4 ^{valence} electrons (Interact upto some extent)



Let's take a piece of silicon & connect it with supply. Now, to flow current there must have free electrons. At 0 K - these Si electrons make a bond with each other, therefore no current flows but at finite temperature some e^- gets free & conduct electricity. Hence at finite temperature (room temp.) silicon also conduct electricity. Therefore it is known as semiconductor. (conducts to some extent).

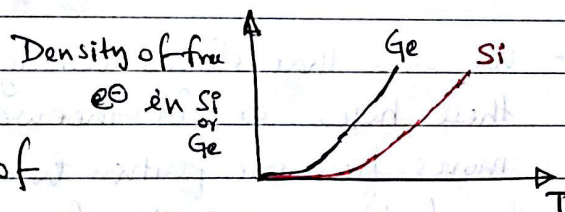
Questions

- ① where do charge carriers come from? ↑ Answer
- ② what types of charge carriers do we have?
- ③ How can we modify the density of charge carriers?
- ④ How do charge carriers move?

Concept of band gap energy:

→ At $T=0K$, No free e^- .

→ When $T \uparrow$ more e^- have a chance to break free because of high thermal energy.



→ Density of $e^- \rightarrow n_i = 5.2 \times 10^{15} T^{3/2} e^{-E_g/2KT}$ $E_g = \text{Band Gap Energy}$
↑
Intrinsic $K = \text{Boltzmann's constant}$
 $= 1.38 \times 10^{-23} \text{ J/K}$

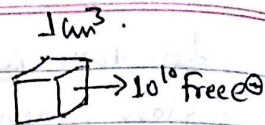
→ The amount of energy required to break the bond of Si atom in order to get free e^- is known as Band Gap Energy (E_g).

→ E_g is higher for Si than Ge bcoz size of Si $<$ Ge. So more energy is required to break the bond. That's why we get more density of e^- in Ge than Si at same temperature.

$$E_g(\text{Si}) = 1.12 \text{ eV}$$

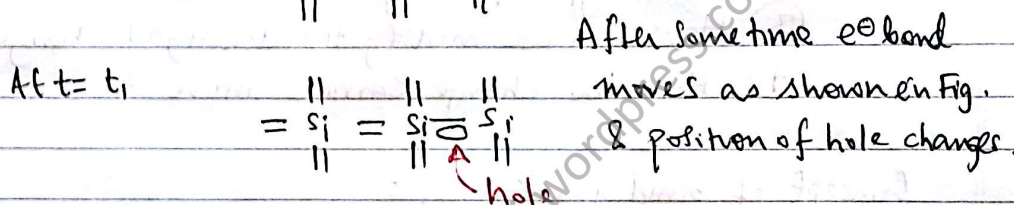
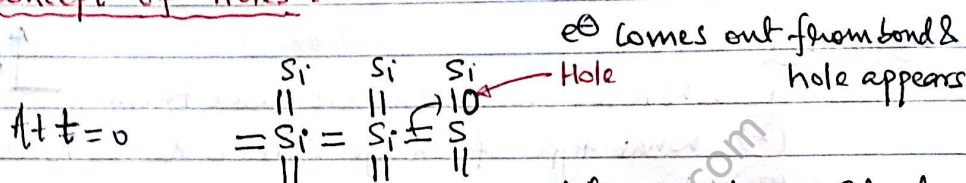
$$E_g(\text{Ge}) = 0.67 \text{ eV}$$

Example: For Silicon at $T=300K$
 No. of free e^- in $\rightarrow \eta_i \approx 10^{10}$ electrons/ cm^3 .
 Pure Silicon/ cm^3 $\approx 5 \times 10^{22}$ Si atoms/ cm^3



- Means very small fraction of atoms released an electron for conduction at room temperature.
- Poor conductor at room temperature.

⊗ Concept of Holes:



- We saw that e^- are coming out from the bond & filling these holes or in other way we can say that hole moves from one position to another. (means +ve charge travel from one position to another). Therefore we can say that holes are capable of conducting current just like e^- .

- Conduction done by $e^- \neq$ conduction done by holes.

Questions:

Why ~~are~~ holes are slower than electron?

- Movement of holes is based on release & trap mechanism & that's why it is slower than the movement of electrons.

- Density of holes = Density of free $e^- = \eta_i$ For Pure Silicon.

$$P = n = \eta_i \Rightarrow Pn = \eta_i^2$$

How do we modify Density of e^- & holes in Si?

We have $n_i = 10^{10}$ electrons/cm³. To increase this amount we do doping.

III	IV	V	
B			
	Si	P	
	Ge.		

*** We add one Phosphorus atom (5 e^- in its outermost shell) So, P shares 4 e^- & one e^- is free to move & participate in Current conduction. So, we don't need to break the bond for conduction. We already have an e^- from P atom for conduction. So, we can make this more conductive by adding more phosphorus in it. This process of adding impurity is known as Doping. Overall this is known as n-type Silicon because it has more no. of free e^- than pure Silicon.

$$\text{Density of P atom} \approx 10^{15} - 10^{17} e^-/\text{cm}^3.$$

∴ No. of free e^- in P atom in 1 cm³ is more than the no. of free e^- in pure Silicon.

→ In Extrinsic Semiconductor, the no. of holes goes down because the free e^- in P atom filled the holes & hence

$$np = n_i^2$$

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classmate

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Lecture: 2

Doping

If we add N donor atoms then density of Free $e^- = N_D / \text{cm}^3$
In Pure Silicon,

No. of Free $e^- = \text{No. of holes.}$

$n =$ density of density of Free e^-

$p =$ density of holes.

For Intrinsic / Pure Silicon : $n = p = n_i \Rightarrow n p = n_i^2$

For Extrinsic / doped Silicon : $n > p$ but $n p = n_i^2$

For doped Silicon (n-type)

$$n \approx N_D$$

$$p = \frac{n_i^2}{n} \approx \frac{n_i^2}{N_D}$$

p is very small

Example



Add 10^{15} Phosphorus atoms

$$n = 10^{15} = N_D / \text{cm}^3$$

$$p = \frac{n_i^2}{N_D} = \frac{10^{20}}{10^{15}} = 10^5 / \text{cm}^3$$

→ By adding 10^{15} Phosphorus atom in 1 cm^3 of Silicon we managed to ↑ no. of free e^- , increases the conductivity to 10^5 times but the no. of holes dropped by 10^5 .

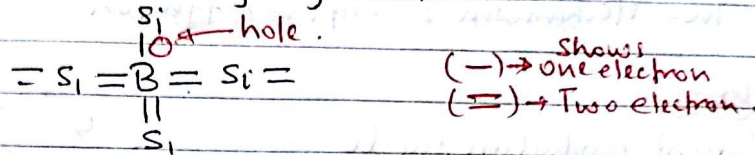
↑
Majority
Carrier

p (in n-type semiconductor)
Minority
Carrier.

No. of
 $N_d \rightarrow$ donor atom / cm^3
 $N_a \rightarrow$ Acceptor atom / cm^3

P-type Semiconductor

Can holes become Majority Carriers?



→ This hole create because Boron does not have an e^- to share. So if we add certain no. of boron atoms, we create that many holes. This is called P-type semiconductor.

→ If we add N_a boron atom / cm^3 then N_a holes / cm^3

$$P = N_a$$

$$n = \frac{n_i^2}{N_a}$$

$$\boxed{np = n_i^2} \text{ always}$$

For doped Silicon (P-type)

$$P = N_a$$

$$n = \frac{n_i^2}{N_a}$$

$$\boxed{P > n}$$

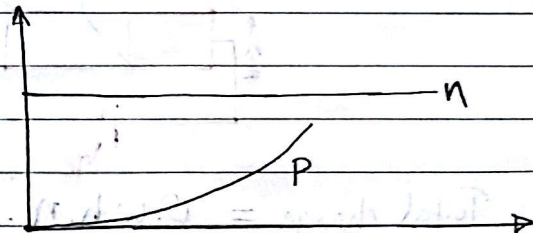
Quiz

What happens to n & p in n-type silicon as temperature increases?

For n-type material

$$n = N_d = \text{constant}$$

$$P = \frac{n_i^2}{N_d}$$



So, At a low of temperature n is approximately constant T but P increases as n_i changes w.r.t T .

SUMMARY :

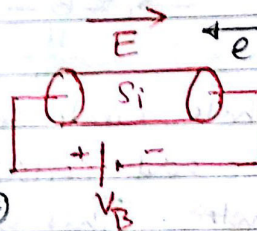
n-type	P-type
(Majority carrier) $n = N_d$	$P = N_a$
(Minority carrier) $P = \frac{n_i^2}{N_d}$	$n = \frac{n_i^2}{N_a}$
$n > P$	$P > n$

How do charge carriers move in semiconductor

Two Mechanism : Drift & Diffusion

Drift

→ Current conduction due to electric field is called Drift

→ When supply is given then e^- accelerates & hits to the atoms sometimes & reaches to terminal velocity (velocity above which e^- can't move because of the resistance offered by an atom).~~Terminal velocity~~

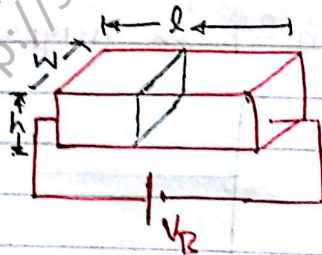
$$v_d \propto E$$

velocity electric field

$$v_d = \mu_n E$$

$$\mu_n = e^- \text{ Mobility}$$

$$= 1350 \text{ cm}^2/\text{Volt-sec.}$$

observation

$$E = V_B / l$$

Total charge = $l \cdot w \cdot h \cdot n \cdot q$ = Total charge passes the cross section

$$I = \frac{\mu_n V_B}{l} w h n q$$

$$= \frac{V_B}{R}$$

$$R = \frac{l}{\mu_n n w h q} = \rho \frac{l}{A}$$

Resistivity of Semiconductor

$$\Rightarrow \begin{cases} A = wh \\ \rho = \frac{1}{\mu_n n q} \end{cases}$$